The Architecture of Learning: Complexity From Simplicity

Near Decomposition is a Fundamental Condition of Learning

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JAN 31, 2026

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One of the most persistent myths in education is that complex skills are irreducibly complex. That reading, writing, problem solving, critical thinking are unified or holistic constructs that resist being broken down into constituent parts and must b absorbed whole or not at all. That if you try to take them apart, you somehow destr the very thing you are trying to cultivate.

This belief is usually framed as humane or naturalistic, a defence of richness agains reduction. This belief is the intellectual foundation of the now widely discredited whole language movement, later rebranded as balanced literacy, with its suspicion decomposition and its faith that immersion in rich texts would allow the parts to organise themselves.

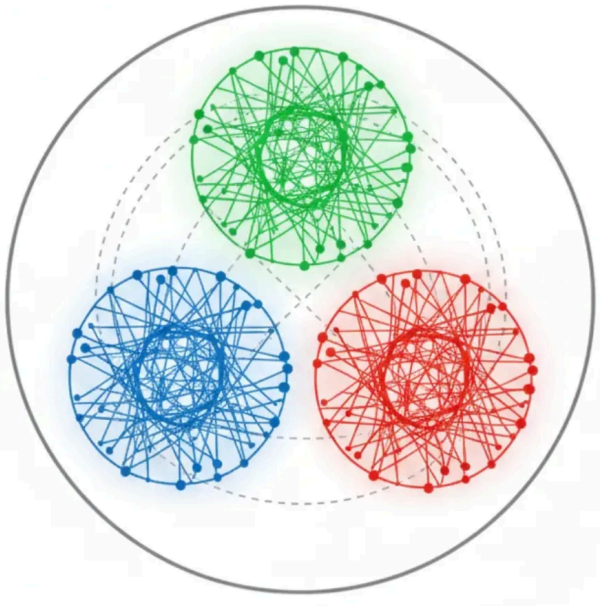
In a paper from 1962, *The Architecture of Complexity*, Herbert Simon set out to explai how complex systems manage to exist at all. His answer was deceptively simple; the do so because they are “nearly decomposable”. In other words, complex systems are not uniform tangles of interdependence. They are clustered and the crucial aspect i that the interactions *within* their component parts are far stronger and more freque than the interactions *between* those parts.

Near Decomposability: The Key Property

Simon illustrates this with a thought experiment. Imagine a building with perfect thermal insulation from the outside. The building is divided into rooms, with good not perfect insulation between them. Each room is further divided into cubicles, wi only poor insulation between cubicles in the same room. If we start with wildly different temperatures throughout the building and wait, what happens?

After a few hours, the cubicles within each room will have reached a common temperature, but there may still be large differences between rooms. After several days, the entire building will approach uniform temperature. The internal equilibration happens fast; the cross-boundary equilibration happens slowly. This temporal separation is the signature of near decomposability.

Simon's point is that in complex systems, parts can be understood and stabilised locally before they must be coordinated globally. In the short run, each subsystem behaves largely independently of the others; in the long run, the system holds toget because those subsystems influence one another only in broad, aggregate ways.



Near Decomposability: Subsystems talk to themselves more than each other.

Put simply, you can understand what is happening inside a subsystem without need to track every detail of what is happening elsewhere. You do not need to model the

entire building to predict how quickly the cubicles in Room 3 will equilibrate. And when you do turn to the building as a whole, you need only the average temperature each room, not the moment by moment fluctuations within them. The hierarchy simplifies. Information compresses. Analysis becomes possible.

The “ill-structured” Domain Problem

This is not merely an observation about thermodynamics. It is a claim about the architecture of anything complex enough to exist and be understood. And it applies directly to the structure of local knowledge and global skills we teach.

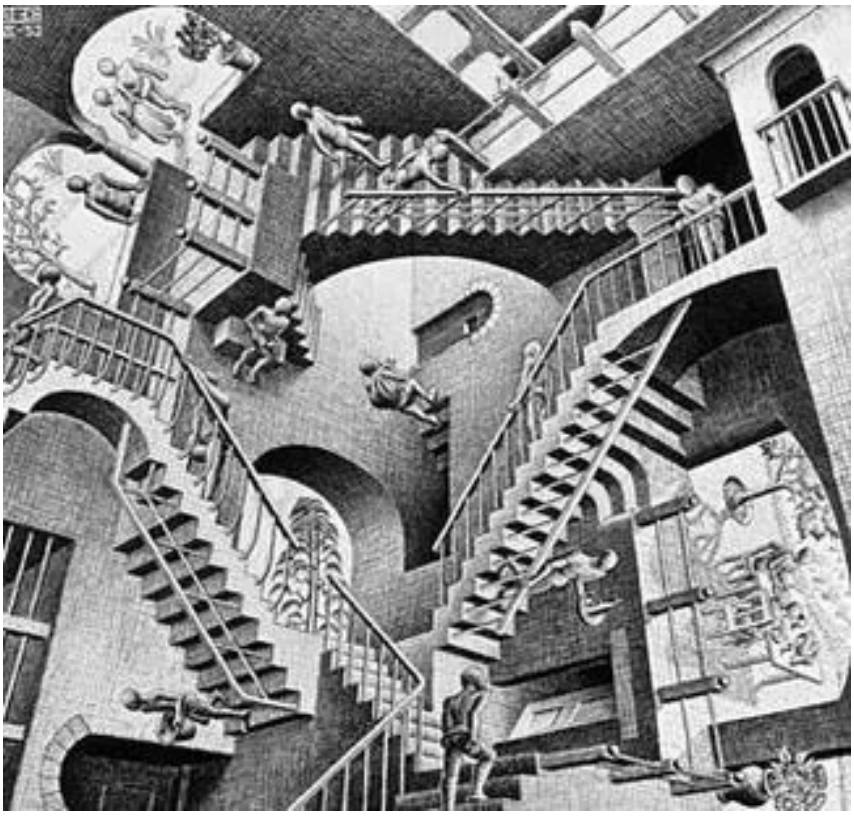
One of my favourite distinctions which emerged out of Artificial Intelligence and Information Processing Psychology in the mid-20th century is the one between a “w defined” or “well structured” domain and an “ill-defined” one. While machines excelled at "well-defined" problems (like chess, where every rule and the win condit are explicit), researchers noticed that many human endeavours (like painting, diplomatic negotiation, or teaching) did not easily fit this framework.

Consider the “ill-defined” domain of reading comprehension. Within the subsysten vocabulary knowledge, interactions are dense and frequent: knowing “benevolent” makes it easier to learn “malevolent”; understanding “chronological” primes you fo “synchronous.” Within the subsystem of domain knowledge, the same pattern holds knowing about the French Revolution makes it easier to understand the Napoleonic Wars; grasping how parliamentary systems work illuminates why coalitions form. These are the cubicles equilibrating rapidly within their rooms.

But the interaction between vocabulary and domain knowledge, while real, is slowe and more aggregate. You do not need to know every word in a history textbook to begin building a schema of nineteenth century Europe, nor do you need a complete model of European history to start acquiring its technical vocabulary. The subsyste can be developed with relative independence in the short run, even though they mu eventually connect.

This is what makes instruction possible. If reading comprehension were not nearly decomposable, if every skill depended on every other skill with equal immediacy, instruction would be hopeless. There would be no place to begin, no component to isolate, no subassembly to stabilise before moving to the next. We could only gestur at the whole and hope.

The curriculum would become an Escher staircase: appearing to ascend while somehow returning to where it began, each step logically connected to the next but whole structure impossible. Students would climb perpetually without rising. Teach would teach perpetually without advancing. The architecture would be elegant, coherent, and utterly untraversable.



Relativity, by M. C. Escher. Lithograph, 1953.

Simon notes that *"for a tolerable description of reality only a tiny fraction of all possible interactions needs to be taken into account"*. This "Empty World" hypothesis is the rea

a teacher can focus on a phonics worksheet without the student needing to underst the global themes of the book simultaneously.

A well designed curriculum respects the architecture of the domain. It identifies th subsystems whose internal interactions are dense, teaches within them until something stable has been built, and only then asks learners to manage the slower, more aggregate connections between them. Phonemic awareness before decoding. Decoding before fluency. Vocabulary and background knowledge developed in parallel, then woven together in increasingly complex texts. The sequence is not arbitrary; it follows from the structure of the skill itself.



Herb Simon

In 1973, Simon returned to the question of complexity, this time asking how proble solvers manage domains that resist formal definition. His answer was characteristically precise. The thing that makes a problem what Simon called "ill structured" is not that it lacks structure altogether, but that its structure must be *discovered and rediscovered* as the solver proceeds.

The architect designing a house, the composer writing a fugue, the teacher plannin unit on the French Revolution: none begins with a complete specification of the

problem space. Constraints emerge. Subgoals are evoked. The problem, as Simon pu it, *"begins to acquire structure by being decomposed into various problems of component design."*

He argued that most ill-structured domains are not ill structured because they lack actual structure, but because their structure *cannot be specified all at once.* The probl space is too large, too interconnected, and too information rich to be processed globally. As a result, structure has to be *constructed locally*, piece by piece, under sev cognitive constraints. In other words, some problems are well structured in the sma and ill structured in the large.

Learning = "Local Structuring"

This matters for learning. Instructional design does not turn ill-structured domains into well structured ones wholesale. It creates temporary islands of structure within them. It narrows the problem space, stabilises a small set of relations, and only then expands outward. Vocabulary, syntax, background knowledge, and inference do not develop simultaneously. They are brought into focus one subsystem at a time, preci because the learner cannot hold the whole system in mind.

Seen this way, near decomposability is not a convenience. It is the mechanism that allows ill structured domains like reading to be learned *at all.* Instruction succeeds when it repeatedly transforms global complexity into locally manageable structure, then reconnects those structures gradually. Without this oscillation between isolati and integration, learning collapses under its own weight.

Again, Simon is not saying that ill structured problems lack structure. He is saying that their structure is *not given in advance.* It has to be constructed during problem solving. At any given moment, the problem solver is working on something well structured: a local task with clear inputs, recognisable operators, and testable outpu It is only the *global* problem that remains ill structured.

Put simply, complex systems can be understood and built because most of the actio happens locally. You can focus on one room without worrying about the whole building, one subsystem without modelling the entire system. Learning works the same way. We make progress by stabilising small, manageable parts first, then coordinating them later. Without this separation of scales, complexity becomes unmanageable and instruction has nowhere to begin.

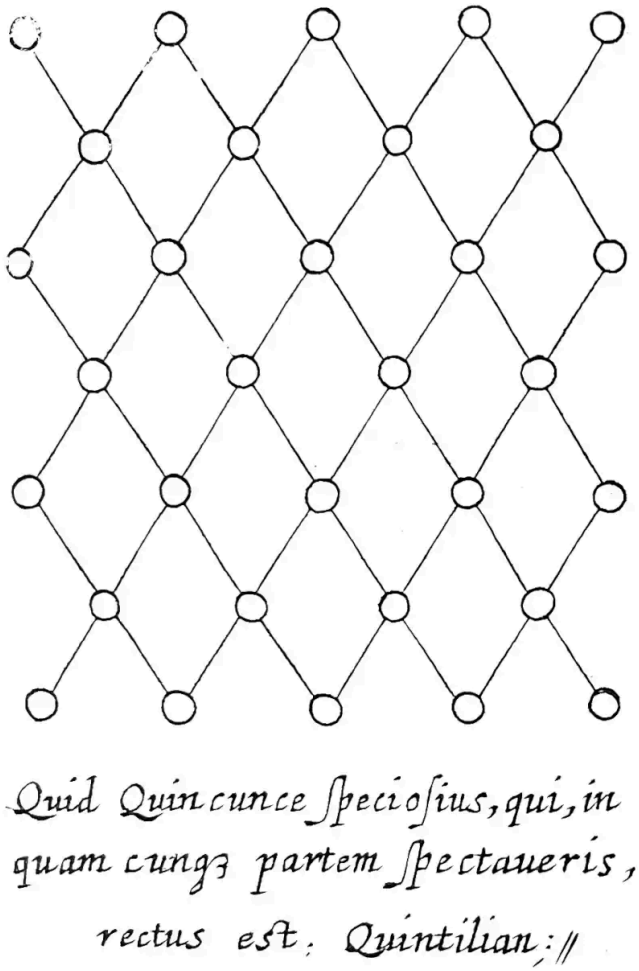
Pattern and Order

Whenever I encounter this topic of how knowledge is structured, I always return to book which had a profound effect on me many years ago; *The Rings of Saturn* by W.G Sebald, in which the narrator, walking the Suffolk coast, finds himself drawn repeatedly to the work of Thomas Browne. Browne, the seventeenth century Norwi physician and polymath, appears throughout the book as a kind of tutelary spirit: a man who saw pattern where others saw only chaos, who traced the same geometric figure, the quincunx, through botany, astronomy, architecture, and scripture. Sebal was fascinated by Browne's conviction that nature is not formless but structured, th the same shapes recur across wildly different domains, that complexity, rightly observed, reveals an underlying order.

> *"All things began in order, so shall they end, and so shall they begin again,"*

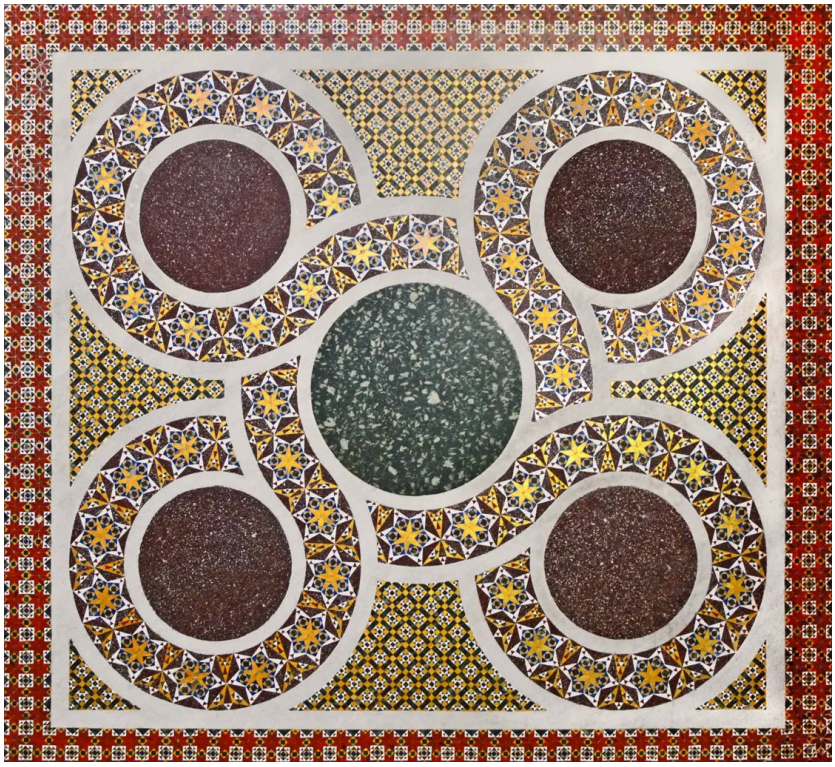
In his 1658 treatise, *The Garden of Cyrus*, Browne offers a mystical and proto-scienti resonance with Simon's idea of near-decomposability. While Simon focused on how complex systems remain stable through hierarchical "sub-assemblies," Browne argu that the entire universe is held together by a single, recurring "sub-assembly": the Quincunx (an arrangement of five points, like the five-dot side of a die).

He was mocked for it, and *The Garden of Cyrus* remains one of the strangest books i English. But the idea is an important one: that complexity is not formless. It has a shape. And the shape recurs.



Frontispiece to *The Garden of Cyrus* (1658). The Latin text, from Quintilian's *Institutio Oratoria*, translates: "What is more beautiful than the quincunx, that, from whatever direction you regard it, presents straight lines?"[1]

Browne was not alone in this intuition. The same conviction, that hidden structures underlie apparent chaos, surfaces again and again across intellectual history. Goeth spent decades searching for what he called the *Urpflanze*, the archetypal plant from which all botanical forms could be derived: a single organisational principle manifesting in endless variation. He never found it literally, but the search led him morphology, the study of how forms transform while preserving deeper structural relationships. Linnaeus had classified plants by surface features; Goethe wanted to understand the grammar beneath the vocabulary



12th-century Cosmatesque mosaic in the Cappella Palatina, Palermo, Sicily

Sebald helps make visible something that Simon formalised and Browne intuited: tl complex understanding grows through patterned recurrence, not total integration from the outset. We do not learn by holding everything in mind at once. We learn b stabilising parts, noticing echoes, and slowly building higher order structure. Seen way, decomposition is not opposed to richness. It is what allows richness to be perceived at all.

The Grain Size Conundrum

Now although this might make *some* sense so far, we have a problem. A big one. If complex skills must be decomposed into smaller components, the question become how far? At what grain-size? at what level of detail should instruction operate?

Too coarse, and the learner is overwhelmed; the subassembly is itself too complex t stabilise, and we have simply reproduced the original problem at a smaller scale. To

fine, and the learner is buried in fragments that never cohere; the parts are so isolat that the connections between them, which give the skill its meaning, are never forn



The Grain-Size Conundrum

I don't have a definitive answer for this right now but what I do know is that it is highly contingent on both the learners prior knowledge and the shape and texture c the knowledge itself. But I think this is ultimately an assessment question to which usually look to Becky Allen who I know has a view on this.

My sense is that the challenge here is to find the level at which the internal interactions within a component are strong enough to create genuine coherence, w the external interactions with other components are weak enough to allow independent practice. And as I say, this is not a fixed property of the domain; it depends on what the learner already knows.

A subassembly that is too large for a novice may be exactly the right size for someo with more prior knowledge, because knowledge itself changes the effective grain si What was once many interacting elements becomes, with sufficient practice, a sing retrievable unit. The expert does not experience the skill as decomposed because, fo them, the parts have fused. The novice needs the decomposition precisely because, them, they have not.

From Chaos to Structure

When a novice enters a domain, almost everything is coupled. Nothing stands still long enough to be grasped. A maths problem feels like a blur of symbols, rules, and decisions. A dense text feels like an uninterrupted stream of unfamiliar words and references. A scientific explanation feels opaque because too many components are interacting at once.

Expertise looks different because the system has been reorganised. Components th once interfered with one another have been stabilised locally. Procedures have becc automatic. Concepts have been compressed into schemas. What was once many interacting elements is now experienced as a smaller number of higher level units.

This is why decomposed knowledge matters so much. Knowledge is what allows pa of the system to become locally stable. It reduces interaction costs. It weakens unnecessary coupling. It turns chaos into structure. Without prior knowledge, every new task is maximally complex because everything depends on everything else. Wit knowledge, complexity becomes manageable because most of the system no longer needs attention.

This is also why the idea that students can simply practise complex skills in their fu form from the outset is so misguided. It ignores the basic architecture of learning. I assumes that the mind can cope with fully coupled systems before any subsystems have equilibrated.

Instruction exists to manage this problem. Good instructional design temporarily isolates subsystems so they can stabilise. It teaches components in ways that allow strong internal connections to form before demanding coordination across boundaries. Phonics before fluency. Sentence construction before extended argume Procedural fluency before strategic choice. These are not arbitrary sequences. They reflect the underlying structure of the domain and the limits of working memory.

What Conditions Need to Be Met For Learning to Happen?

So if near decomposability is the architecture that makes complex learning possible then certain principles of instruction are not mere pedagogical preferences. They a invariants: conditions that must be met if learning is to occur at all. Two follow directly from Simon's analysis.

First: subsystems must be stabilised before they are coordinated. If the interactio within a component are stronger and more frequent than the interactions between components, then those internal interactions must be allowed to equilibrate first. T cubicles must reach a common temperature before the rooms can; the rooms must stabilise before the building does. Skip this sequence and the system never settles.

In instruction, the implication is immediate. You cannot ask learners to coordinate subsystems that have not yet stabilised internally. Teaching decoding and comprehension strategy simultaneously, before either is secure, is not ambitious teaching; it is architecturally incoherent. The learner is being asked to manage cros boundary interactions while the boundaries themselves are still in flux.

The violation signatures are predictable. Performance collapses as elements combi Error rates spike at integration points. Progress stalls despite continued exposure. Learners appear to "know" components in isolation but cannot use them together. These are not motivational failures or signs of limited ability. They are diagnostic signals that coordination has been demanded before stabilisation was achieved.

Second: any decomposition must support recombination. The goal is not to shatte skill into permanent fragments but to isolate components long enough for them to stabilise, then reconnect them. A curriculum that teaches phonics in perpetual isolation, never asking learners to read connected text, violates the architecture jus surely as one that skips phonics entirely. The oscillation between local stability and global coordination is not optional. It is how complex skill is assembled.

The question for any instructional design then becomes structural: does it allow subsystems to stabilise before demanding coordination? Does it reconnect what it l separated? If not, the design will fail, not because it offends tradition, but because i violates the conditions under which complex things can be learned at all.

If Learning is to Fail, It Needs to Fail Locally Not Globally

There is also a third condition that needs to be met if the structure is to be reliably built. Simon's genius was in realising that for a system to evolve, or for an organism learn, it must be able to fail *locally without failing globally*. This is the deep insight buried in the architecture of complexity: error must be containable and the reason t this is so that it can be fixed before everything else falls apart, so that remediation c occur swiftly.

Consider a child who cannot yet decode fully being asked to "read for meaning." W comprehension fails, as it must, where is the error? In the decoding? The vocabular The background knowledge? The inference? The failure is everywhere and therefor nowhere. But if decoding is isolated and practised until stable, errors in decoding c be seen, corrected, and resolved before they contaminate everything else.

When instruction respects near decomposability, failure becomes local. The studen practising sentence construction fails at sentence construction. The error is specifi immediate, and actionable. She can see what went wrong, adjust, and try again. The subassembly can be rebuilt without dismantling everything else. And because failu local, iteration is fast.

This is why complex systems can evolve at all. Simon illustrated the point with his parable of the two watchmakers, Hora and Tempus. Both made watches with a thousand parts. Tempus built his watches as single assemblies; if he was interrupte any point, the entire structure fell apart and he had to start again. Hora built his watches from stable subassemblies of ten parts each, which combined into larger

subassemblies, which combined into the whole. When Hora was interrupted, he los only the subassembly in progress. Tempus went bankrupt. Hora flourished. The difference was not skill or effort. It was architecture.

Learning is the construction of complex cognitive systems. It obeys the same logic. When a novice attempts a complex skill in its full form, failure is global. The studer who tries to write an essay before mastering sentences does not receive useful feedback about sentences; she receives feedback about essays, which is too distant, aggregate, too diffuse to guide improvement. The error signal cannot be localised. Everything is wrong, which means nothing specific can be fixed.

The skills we most value; writing, reading comprehension, mathematical reasoning scientific thinking, are not mystical wholes that emerge when conditions are favourable. They are systems of subassemblies with strong internal connections anc weaker external ones. They can be built, but only if we respect the architecture that makes building possible. Only if we allow the parts to stabilise before demanding coordination. Only if we let failure happen where it can be contained, corrected, an converted into progress.

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Skip Gole Robert Gole 2d

Your description of an "architecture that makes building possible" and how "Only if we allow t parts to stabilise before demanding coordination" brought back memories of using dictogloss language learners at varying levels. This approach uses a dictation-type activity in 3-4 rounds first step is traditional dictation where students write down what they heard. The teacher doe speak slowly, but at a natural rate. Second, they create groups and use their transcribed word work together to fix their text. The idea is to create a better, more clear, textual version. They again and rewrite once more. Finally, the activity changes to whole class discussion with all gr writing a final version on the board. Guided by the teacher, they create a best version, which i chosen by all. There is constant failure, but it is group-oriented with teacher guidance (monit so mistakes "can be contained, corrected, and converted into progress." For many steps, the correcting is done by them, so they get a good gloss on the 5-7 sentences. In fact, there is a deal of interaction because of the focus on making its meaning clear. This is key to the whole lesson. If you choose more than 5-7 ideas, it might overwhelm by making the focus to 'fine' fo level instead of 'granular'. This all depends on the language level of the learners. I've done thi beginning level to advanced language learners.

Thanks for this thought piece. It's focused my thoughts on many memories from my teaching

LIKE (2) REPLY

1 reply

Andrew Evans Andrew Evans 2d

From Men are Different, by Alan Bloch:

"Our scientists tell us that Men were very much like us -- and the skeleton of a Man is, to be s almost the same as the skeleton of a Robot, except that it's made of some calcium compound instead of titanium. Just the same, there are other differences.

"It was on my last field trip, to one of the inner planets, that I met the Man. He must have beer last Man in this system, and he'd forgotten how to talk -- he'd been alone so long. Once he le

our language, we got along just fine together, and I planned to bring him back with me. Some
happened to him, though.

"One day, for no reason at all, he complained of the heat. I checked his temperature and deci
that his thermostat circuits were shot. I had a kit of field spares with me, and he was obvious
of order, so I went to work. I turned him off without any trouble. I pushed the needle into his n
operate the cutoff switch, and he stopped moving, just like a Robot. But when I opened him u
wasn't the same inside. And when I put him back together, I couldn't get him running again. T
sort of weathered away -- and by the time I was ready to come home, about a year later, there
nothing left of him but bones. Yes, Men are indeed different."

LIKE (2) REPLY

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